Does Diplomatic Network Size Increase UNGA Coalition Size?

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Introduction

Diplomatic posts, such as embassies, consulates, permanent missions, and high commissions, function as the operational nodes of a country's diplomatic network abroad. The main purposes of these establishments include protecting/serving their national citizens abroad, conducting bilateral negotiations, gathering political intelligence, and soft power through representation.[1] On average, governments spend millions to billions of dollars annually to maintain their diplomatic infrastructure abroad, depending on the size of their network.[2] Thus, it is important to investigate the effectiveness of these posts in producing the desired diplomatic relations, or the marginal effect of having an additional post abroad. The research question of this paper asks whether countries' size in their diplomatic network correlates with their coalition size in the UNGA Resolution votes. The premise was tested through statistical analysis on the correlation between the number of diplomatic posts and the number of United Nations General Assembly's Resolution Votes that align with each study country. The results from the main regression model suggest that, in cross-sectional comparison, the number of diplomatic posts is

[1] Vienna Convention on Diplomatic Relations. (1961, April 18). *United Nations Treaty Series, 500*, 95. https://legal.un.org/ilc/texts/instruments/english/conventions/9_1_1961.pdf

[2] U.S. Department of State. (2024). *Congressional budget justification: Department of State, foreign operations, and related programs FY 2025*. https://www.state.gov/wp-content/uploads/2024/03/FY-2025-CBJ-Final.pdf

negatively correlated with the amount of votes coalition countries can acquire. Alternative models with adjusted parameters still oppose the theory that diplomatic posts increase countries' diplomatic influence in international forums.

Theoretical Framework and Hypothesis

The main hypothesis is that an increase in diplomatic posts would lead to positive bilateral relations with host countries and benefit sending countries' international diplomacy extensively. Hence, when acting in international forums, such investments would lead to positive alignment with countries. The United Nations General Assembly provides an ideal field study, with 193 member states voting on hundreds of resolutions annually; it is one of the most significant multilateral forums acknowledged globally. Thus, the theory assumes that diplomatic investment should be detectable if it were to manifest multilateral influence anywhere. Prior work demonstrated that diplomatic network sizes are structural outcomes driven by state power, geographic proximity, and ideological alignment rather than randomly distributed investments.[3] Additionally, bilateral diplomatic ties function as credible signals of cooperative intent,[4] and the use of UNGA voting alignment as a proxy for foreign policy preferences already exists.[5] This paper tests the previous theories and methodologies through the framework of using diplomatic posts as a proxy to measure the effectiveness of countries' diplomatic networks with UNGA vote alignment as the outcome.

Methodology and Empirical Analysis

[3] Neumayer, E. (2008). Distance, power and ideology: Diplomatic representation in a world of nation-states. *Area, 40*(2), 228–236. https://doi.org/10.1111/j.1475-4762.2008.00811.x
[4] Kinne, B. J. (2014). Dependent diplomacy: Signaling, strategy, and prestige in the diplomatic network. *International Studies Quarterly, 58*(2), 247–259. https://doi.org/10.1111/isqu.12047
[5] Bailey, M., Strezhnev, A., & Voeten, E. (2017). Estimating dynamic state preferences from United Nations voting data. *Journal of Conflict Resolution, 61*(2), 430–456. https://doi.org/10.1177/0022002715595700

Three datasets were used for this paper: The Global Diplomacy Index from the Lowy Institute, UNGA resolution votes from the UN Digital Library, and the World Development Index. The primary driver is the Global Diplomacy Index, which documents country rankings based on the total number of diplomatic posts each country has sent abroad. The number of diplomatic posts is the independent variable, covering the top 60 most diplomatically active states in 2016, 2017, 2019, 2021, and 2023.

The UNGA resolution votes from the UN Digital Library provide the voting records and were filtered to the five Lowy waves. The dependent variable, coalition size, was created by filtering to contested resolutions defined as those receiving at least five yes votes and at least five no votes from UN members to exclude unanimous resolutions, resulting in a total of 237 resolutions. A dyadic agreement score from contested resolutions was calculated by adding identical votes countries held from the resolution results, and was divided by the total number of contested resolutions each pair of countries voted in. Finally, coalition size was then defined as the sum of one country's dyadic agreement scores with other paired countries that were above a 60% threshold in a given year, meaning the total number of other countries with whom a focal country achieved an agreement score exceeding 0.60 in a given year.

Finally, GDP per Capita, trade openness as a percentage of GDP, and population were taken from the WDI to control for countries' socioeconomic status that could potentially influence alignment from resolution votes beyond the effect of diplomatic posts. GDP per Capita and population were log-transformed to address the skewness of the sample between large and small countries. Lastly, Observations with missing values on any model variable were dropped, primarily driven by 15 missing values in Trade Openness, resulting in a final sample of 275 country-year observations as the unit of analysis.

Exploratory Data Analysis

The exploratory data analysis figures for the independent and dependent variables are shown in the appendix. Figure 1 shows the distribution of diplomatic network size from all country-year observations within the sample. The figure shows that although the sample mean is around 121.2 posts, the distribution is slightly right-skewed with a spike of observations between 30-45 posts, signaling that the sample contains a considerable number of countries with relatively small diplomatic networks. Figure 2 shows the distribution of coalition size, with one larger cluster around 45-50 coalition size, and a second around 120- 140, while the sample mean is 77.1 coalition size. Figure 3 presents the bivariate relationship between diplomatic network size and coalition size for 2023, fitting the bivariate regression line to show the two variables' relations. The figure shows a negative correlation between diplomatic posts and coalition size, suggesting that countries with a larger diplomatic network actually have a smaller coalition size in the UNGA resolution votes in 2023.

Statistical Analysis

The main regression model sets UNGA Coalition Size as the dependent variable, Diplomatic Posts as the independent variable, and log GDP per Capita, Trade Openness, and log Population as the control variables.

$$\boxed{\text{CoalitionSize}_{it} = \beta_0 + \beta_1\,\text{DiplomaticPosts}_{it} + \beta_2\ln(\text{GDPpc}_{it}) + \beta_3\,\text{TradeOpen}_{it} + \beta_4\ln(\text{Pop}_{it}) + \varepsilon_{it}}$$

The main Pooled OLS model (M1) shows that across the country-year observations, each additional increase in diplomatic posts is correlated with a -0.319 decrease in coalition size at the 60% threshold, statistically significant at the 0.001 level. Additionally, a percentage increase in GDP per Capita is also significantly correlated with coalition size at the 0.001 level, with the coefficient at -17.273. In contrast, a percentage increase in population is correlated with a 10.466

increase in coalition size, also at the 0.001 level. Lastly, the trade openness coefficient suggests a positive correlation of 0.037 with coalition size, but shows no statistical significance. The R^2 of the regression model is 0.573, suggesting that the variables within the model were able to explain a decent proportion of the variance. From the M1 model, the results strongly opposed the hypothesis that a larger number of diplomatic posts leads to a larger coalition size within the UNGA resolution vote alignments.

Additionally, alternative models were tested to assess the robustness of the main finding. The second model (M2) in Table 1 added year fixed effects to the M1 model to control for the global time trend that all countries experience in a given year. M2 results had little difference from the M1 model, suggesting the negative relationship is not driven by secular trends over time. The third model (M3) added country fixed effects to compare each country to itself across waves rather than to other countries. The diplomatic posts coefficient in M3 turned to 0.154 and lost statistical significance, indicating that expansion of diplomatic networks over time does not cause measurable changes in coalition size. The R^2 of M3 increased to 0.977, suggesting that coalition alignment is largely determined by within-country time-invariant effects rather than shifts in diplomatic network size, country wealth, or trade openness over time.

Predicted coalition size for M1 was computed with all control variables held at their sample means. Figure 4 shows the predicted coalition size decreasing consistently as diplomatic network size increases, with a 95% confidence interval indicating statistical reliability across the range. The first difference was calculated by comparing the 25th percentile of diplomatic posts against the 75th percentile, which showed a difference of −24.4 coalition partners. This means that a country at the higher end of the diplomatic network distribution is predicted to align with

approximately 24 fewer UN member states than a country at the lower end, holding wealth, trade openness, and population constant.

Alternative parameters on coalition size thresholds, outlier exclusion, and contested resolution definitions were also tested to assess the sensitivity of the main finding. Table 2 estimated the main model across five coalition size thresholds, 0.30, 0.50, 0.60, 0.70, and 0.90 to test whether the cutoff in coalition threshold would alter the coefficients. The results show that the negative coefficient on diplomatic posts remains statistically significant at the 0.001 level across all specifications except the broadest threshold (0.30). Table 3 tests outlier sensitivity by identifying the six most influential countries through Cook's Distance to remove from the model. Leaving out Russia, Singapore, Brunei, Saudi Arabia, Turkey, and Australia. The coefficient on diplomatic posts changes from −0.318 in the full sample to −0.165 when the outliers were excluded, but it still remained negative and significant at the 0.001 level. This suggested that the negative coefficient was universal across all country-year observations in the sample.

Table 4 tested coalition size as a proportion by dividing the coalition size by the total UN membership (193 countries), converting to a percentage-point measure across the five thresholds of 0.30, 0.50, 0.60, 0.70, and 0.90. Similar to the results from Table 2, the coefficient on the diplomatic posts remained negative and significant except for the 0.30 threshold, which confirmed that the result is not dubious to how coalition size is scaled. Lastly, Table 5 tested alternative definitions of contested resolutions with different thresholds of requiring at least ten non-alignment votes, at least 20% nonalignment votes, and non-unanimous that only excluded resolutions where all countries agreed. The diplomatic posts coefficient for Table 5 remained negative and significant at the 1% level across all four resolution rules, confirming that the results are also not sensitive to how contested resolutions are defined.

Discussion & Limitations

Several limitations constrained the scope and causal interpretation of the results from the models. First, the sample size is significantly constrained by the available observations from the Lowy Institute Global Diplomacy Index. The index covers only the 60 most diplomatically active states, excluding the rest of the UN members that maintain smaller and less globally distributed networks. The relationship between diplomatic investment and coalition size could potentially look different for smaller developing states. Additionally, the index is available in only five waves, which limited within-country variation for the panel data observations. Additionally, the data could potentially be capturing countries that have already developed and invested their diplomatic networks extensively, where the marginal benefits of adding additional posts are already diminishing. Thus, the coefficients might show a different pattern if the data included observations from earlier periods of diplomatic network expansion.

Second, the independent variable treated all diplomatic posts equally, regardless of size, staffing, capacity, or strategic importance. This binary documentation says nothing about post quality, diplomatic effectiveness, or the resources devoted to multilateral lobbying specifically. A country that closes twenty small consulates and opens five large strategically positioned embassies would show a net decline in posts while potentially increasing its diplomatic influence substantially. The incorporation of capacity, budget allocations, or staffing levels could improve measurement validity. Furthermore, the study treated vote alignment as the primary purpose of diplomatic posts, while other functions, such as negotiating ground, citizen accommodation abroad, and intelligence sourcing, might be more effective in justifying the expansion of countries' diplomatic networks.

Third, the dependent variable assumes that UNGA voting alignment is the direct, detectable result of diplomatic coalition-building, which may not have fully captured how coalitions actually operate in international politics. Coalition partners could potentially coordinate through abstention rather than an identical vote, and thus be omitted from the study through the dyadic agreement score framework. Additionally, diplomatic coalitions can manifest in other domains that are outside the scope of UNGA voting, such as trade agreements, security partnerships, foreign aid relationships, and informal bilateral coordination that would be undertaken outside of the international forum. Thus, a country could theoretically have a highly effective diplomatic coalition that is simply invisible in UNGA voting data. Ultimately, future studies should account for these limitations and provide alternative methods to fully validate the findings.

Conclusion

The results across all models suggest that the size of the diplomatic network does not benefit countries in gaining more partners to align their votes with the sending countries within the UNGA resolutions. This suggests that countries should carefully consider whether to allocate more of their national budget to building additional diplomatic posts, especially for the purpose of obtaining more voting coalitions in the UNGA. Diplomatic posts may still be justified on bilateral, commercial, and consular grounds, but coalition-building at the UNGA requires alignment in underlying foreign policy positions that physical diplomatic presence alone cannot produce.

References

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World Bank (2024). World Development Indicators [Dataset].

Appendix

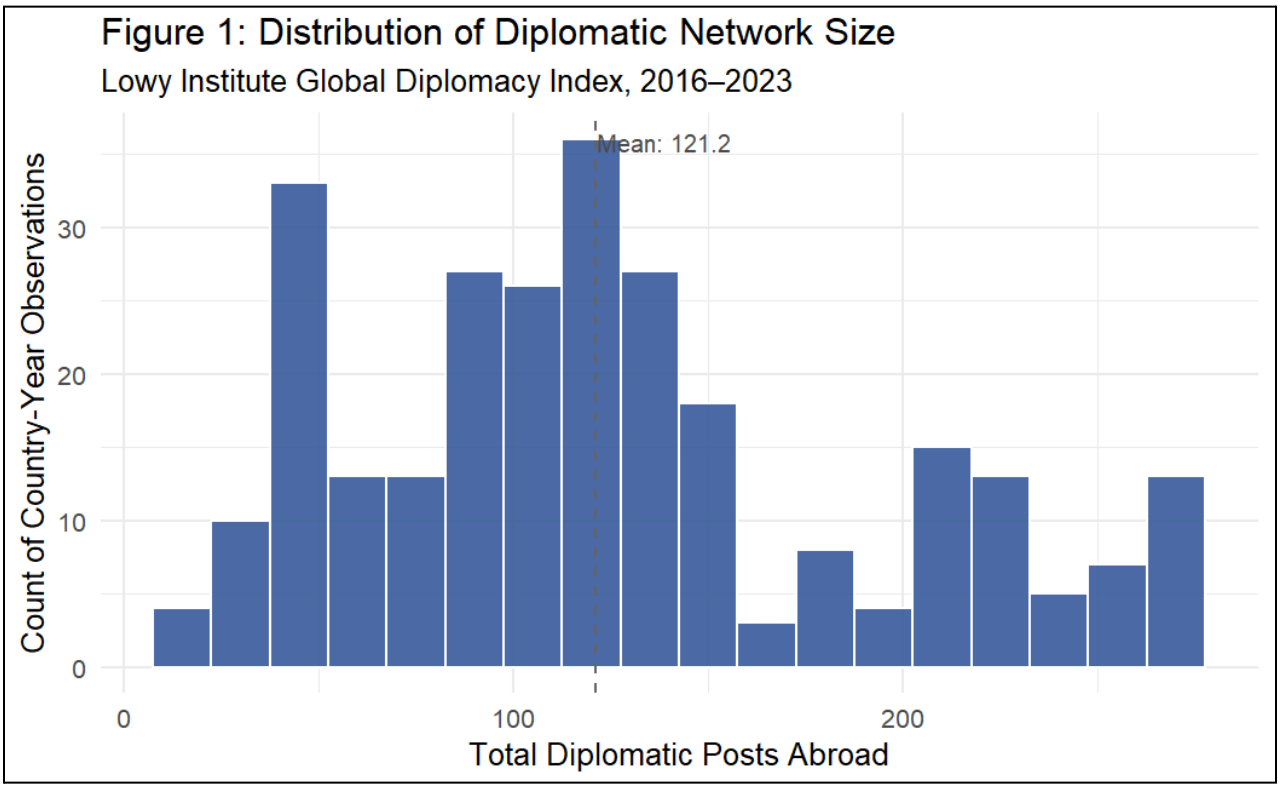

Figure 1. Distribution of total diplomatic posts abroad across all country-year observations (N = 260). Data from the Lowy Institute Global Diplomacy Index across five waves: 2016, 2017, 2019, 2021, and 2023. The dashed line indicates the sample mean (121.2 posts). The distribution is slightly right-skewed, with the largest concentration of observations between 30-45 posts.

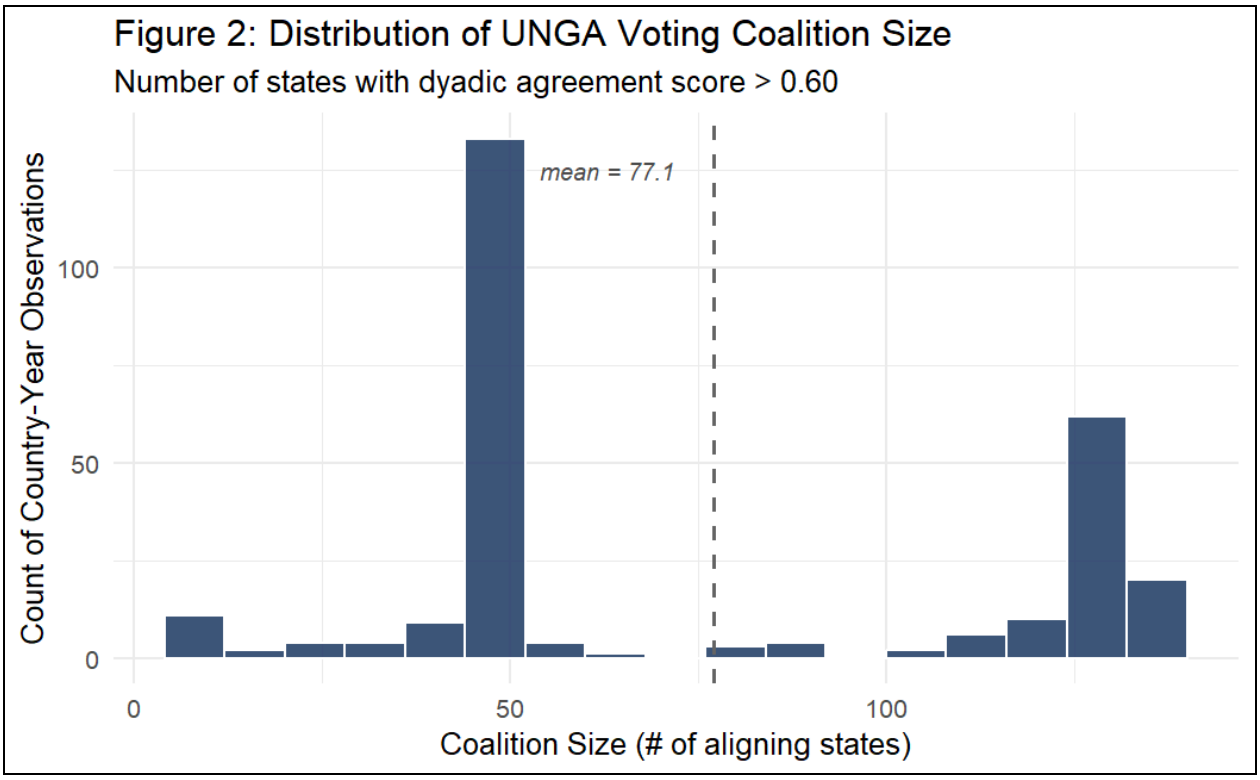

Figure 2. Distribution of UNGA voting coalition size across all country-year observations (N = 260). Coalition size is defined as the number of UN member states with a dyadic voting agreement score exceeding 0.60 in a given year. The dashed line indicates the sample mean (77.1 states). The distribution is bimodal, with one cluster concentrated around 45-50 and a second cluster above 120.

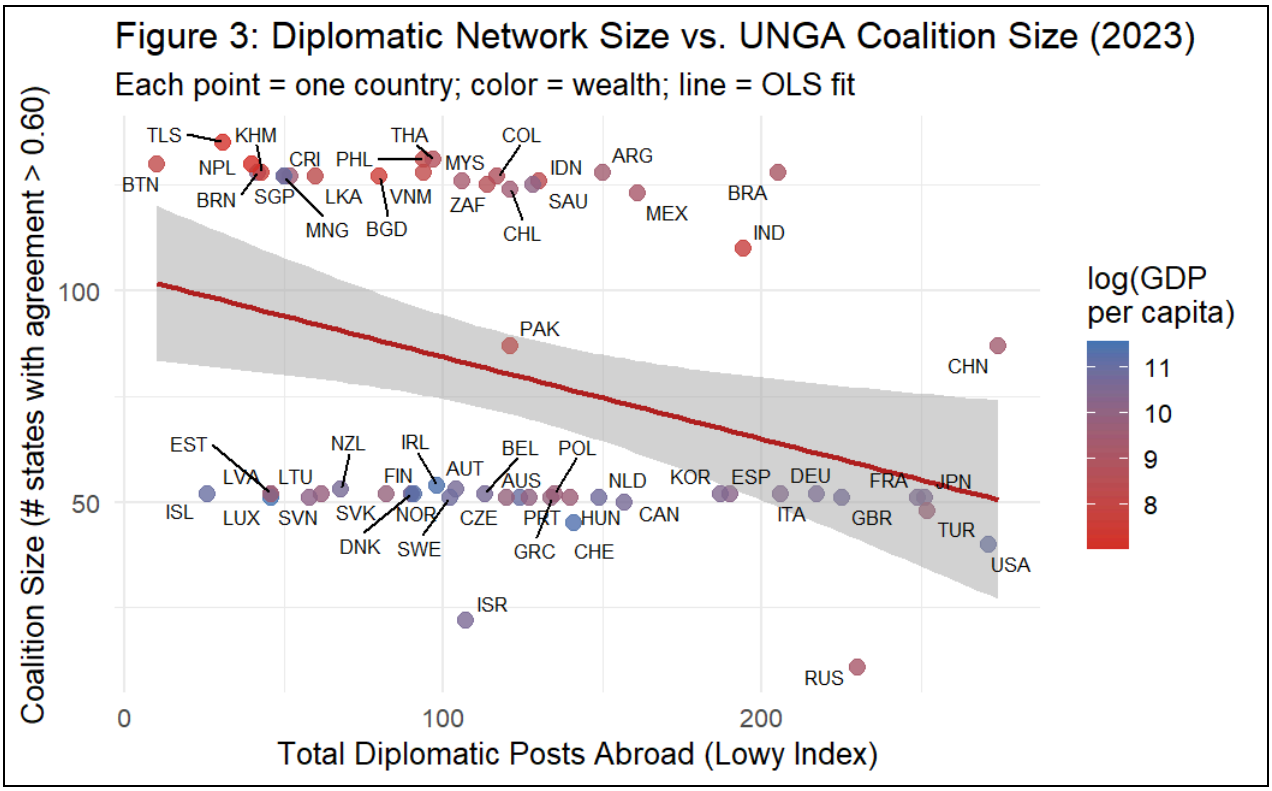

Figure 3. Bivariate scatter plot of total diplomatic posts abroad (x-axis) against UNGA coalition size (y-axis) for 2023 (N = 60). Each point represents one country. Point color indicates log GDP per capita, with red indicating lower wealth and blue indicating higher wealth. The red line shows the OLS fit with 95% confidence interval. The negative slope indicates that countries with larger diplomatic networks tend to have fewer coalition partners.

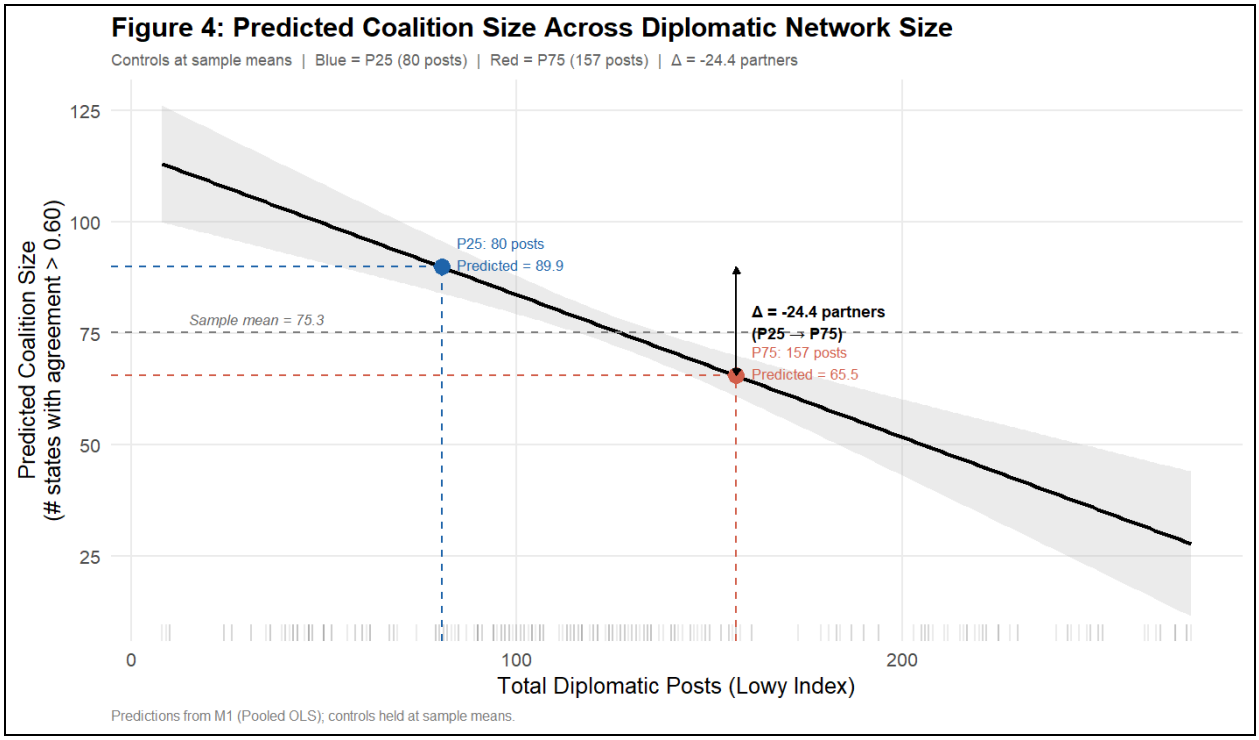

Figure 4. Predicted UNGA coalition size across levels of diplomatic network size, with all control variables held at sample means. Predictions from M1 (Pooled OLS). The blue point shows the predicted coalition size at the 25th percentile of diplomatic posts (80 posts = 89.9 partners) and the red point at the 75th percentile (157 posts = 65.5 partners), yielding a first difference of −24.4 coalition partners. The dashed horizontal line indicates the sample mean coalition size (75.3 states).

Table 1: Diplomatic Network Size and UNGA Coalition Size

	M1: Pooled OLS	**M2: Year FE**	**M3: Two-way FE**
Panel A: Coefficients			
Diplomatic Posts	−0.319***	−0.318***	0.154
	(0.054)	(0.026)	(0.127)
log(GDP per Capita)	−17.279***	−17.273***	−22.825*
	(2.282)	(1.027)	(13.226)
Trade Openness	0.037	0.036	−0.018
	(0.031)	(0.010)	(0.074)
log(Population)	10.466***	10.436***	−13.093
	(2.084)	(1.257)	(8.558)
Constant	248.958***		
	(21.761)		
Panel B: Model Fit			
Year FE	No	Yes	Yes
Country FE	No	No	Yes
Observations	275	275	275
R^2	0.573	0.575	0.977
Adj. R^2	0.566	0.562	0.970

Note: Standard errors in parentheses.

DV = coalition size (# states with dyadic agreement > 0.60).

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 2: Robustness — Alternative Agreement Thresholds

	0.30 (Broad)	**0.50 (Moderate)**	**0.60 (Main)**	**0.70 (Strong)**	**0.90 (Near-perfect)**
Panel A: Coefficients					
Diplomatic Posts	−0.084	−0.303***	−0.319***	−0.307***	−0.121***
	(0.076)	(0.061)	(0.055)	(0.054)	(0.036)
log(GDP per Capita)	−10.522***	−15.859***	−17.279***	−16.350***	−8.699***
	(3.196)	(2.558)	(2.282)	(2.417)	(1.497)
Trade Openness	0.002	0.016	0.037	0.032	0.043**
	(0.043)	(0.034)	(0.031)	(0.031)	(0.020)
log(Population)	1.117	10.756***	10.466***	6.970***	1.322
	(2.918)	(2.336)	(2.084)	(2.207)	(1.367)
Constant	255.177***	245.982***	238.079***		112.236***
	(30.472)	(24.388)	(23.046)		(14.275)
Panel B: Model Fit					
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	275	275	275	275	275
R^2	0.124	0.492	0.573	0.503	0.373
Adj. R^2	0.111	0.485	0.566	0.495	0.364

Note: Standard errors in parentheses.
DV = number of states with dyadic agreement score above threshold.
0.30 = broad alignment; 0.50 = moderate; 0.60 = main specification;
0.70 = strong alignment; 0.90 = near-perfect alignment.
Year FE included throughout. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 3: Robustness — Excluding Influential Observations

	Main (M2)	Excl. All 6 Countries
Diplomatic Posts	-0.318***	-0.165***
	(0.055)	(0.058)
log(GDP per Capita)	-17.273***	-22.464***
	(2.302)	(2.330)
Trade Openness	0.036	0.006
	(0.031)	(0.030)
log(Population)	10.436***	6.394***
	(2.097)	(2.130)
Num.Obs.	275	248
R2	0.575	0.676

* p < 0.1, ** p < 0.05, *** p < 0.01

All 6 = Russia, Singapore, Brunei, Saudi Arabia, Turkey, Australia. Year FE included. * p<0.10, ** p<0.05, *** p<0.01

Table 4: Robustness — Proportion DV Across Agreement Thresholds

	0.30 (Broad)	**0.50 (Moderate)**	**0.60 (Main)**	**0.70 (Strong)**	**0.90 (Near-perfect)**
Panel A: Coefficients					
Diplomatic Posts	−0.052	−0.159***	−0.165***	−0.160***	−0.064***
	(0.037)	(0.032)	(0.028)	(0.030)	(0.018)
log(GDP per Capita)	−6.013***	−8.455***	−8.950***	−8.478***	−4.515***
	(1.538)	(1.326)	(1.193)	(1.259)	(0.771)
Trade Openness	0.011	0.011	0.019	0.018	0.024**
	(0.021)	(0.018)	(0.016)	(0.017)	(0.010)
log(Population)	0.870	5.624***	5.407***	3.675***	0.737
	(1.401)	(1.208)	(1.086)	(1.147)	(0.702)
Panel B: Model Fit					
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	275	275	275	275	275
R^2	0.261	0.503	0.575	0.508	0.394
Adj. R^2	0.238	0.488	0.562	0.493	0.376

Note: Standard errors in parentheses.

DV = (coalition size / 193) × 100, expressed as % of UN members.

Coefficients = percentage point change per additional diplomatic post.

0.60 is the main specification. Year FE included throughout.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Table 5: Robustness — Alternative Contested Resolution Definitions

	R1: ≥5/≥5 (Main)	**R2: ≥10/≥10 (Stricter)**	**R3: Minority ≥20%**	**R4: Non-Unanimous**
Panel A: Coefficients				
Diplomatic Posts	−0.319***	−0.384***	−0.320***	−0.345***
	(0.054)	(0.055)	(0.051)	(0.072)
log(GDP per Capita)	−17.279***	−12.602***	−12.231***	−13.510***
	(2.282)	(2.303)	(2.153)	(3.014)
Trade Openness	0.037	−0.006	0.015	−0.003
	(0.031)	(0.031)	(0.029)	(0.041)
log(Population)	10.466***	10.186***	11.088***	11.022***
	(2.084)	(2.103)	(1.966)	(2.752)
Constant	248.958***	213.644***	199.899***	238.203***
	(21.761)	(21.962)	(20.532)	(28.737)
Panel B: Model Fit				
Fixed Effects	None	None	None	None
Observations	275	275	275	275
R^2	0.573	0.523	0.519	0.392
Adj. R^2	0.566	0.516	0.512	0.383

Note: Standard errors in parentheses.

DV = coalition size (agreement > 0.60) under each resolution rule.

R1: ≥5 yes AND ≥5 no votes (main specification).

R2: ≥10 yes AND ≥10 no votes (stricter).

R3: minority side receives ≥20% of total votes cast.

R4: excludes only unanimous resolutions.

Pooled OLS, no fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$